

Lecture 5:

Set Theory

Basic Sets

- Unordered collection of objects without duplicates
- $\{1,3,5\} = \{1,5,3\}$
- {"tire", "axel", "lever"}
- {Hilbert, Apollo, Archimedes}
- $\{\} = \emptyset$

Infinite Sets

- $\mathbb{N}$ = Natural numbers = $\{0, 1, 2, \dots\}$
- $\mathbb{Z}$ = Integers = $\{0, 1, -1, 2, -2, \dots\}$
- $\mathbb{R}$ = Real numbers
- $\mathbb{Q}$ = Rational Numbers

Set Comprehensions

- $\mathbb{Q} = \{a/b \mid a \text{ and } b \text{ are numbers and } a \text{ and } b \text{ are relatively prime}\}$
- $C = \{x \mid C x\}$
- $F = \{x \mid F x \text{ elvis}\}$
- $FS = \{x \mid F x \text{ elvis} \wedge S x\}$

Set Relations

- $\in$ (in)
- $\subset$
- $1 \in \{1,3,5\}$
- $2 \notin \{1,3,5\}$
- $\{1,3\} \subset \{1,3,5\}$
- $\neg (\{1,2\} \subset \{1,3,5\})$
- $\{1\} \notin \{1,3,5\}$
- NOT $3 \subset \{1,3,5\}$
- $\{1,3,5\} \subseteq \{1,3,5\}$

Set Operations

- Union: $\cup$
 - $\{1,3,5\} \cup \{0,1,2,3\} = \{0,1,2,3,5\}$
 - $\{x : C x\} \cup \{x : F x\} = \{x : C x \vee F x\}$
 - $A \cup B = \{x \mid x \in A \vee x \in B\}$
- Intersection: $\cap$
 - $\{1,3,5\} \cap \{0,1,2,3\} = \{1,3\}$
 - $\{x : C x\} \cap \{x : F x\} = \{x : C x \wedge F x\}$
 - $A \cap B = \{x \mid x \in A \wedge x \in B\}$
- Set Difference: $\setminus$
 - $\{1,3,5\} \setminus \{0,1,2,3\} = \{5\}$
 - $A \setminus B = \{x : x \in A \wedge x \notin B\}$
- $\bar{A} = \{x : x \notin A\}$

$\bar{A}$

Powersets

- $P(A)$ = set of all subsets of A
- 2^A
- $P(\{1,3,5\}) = \{\emptyset, \{1\}, \{3\}, \{5\}, \{1,3\}, \{1,5\}, \{3,5\}, \{1,3,5\}\}$

Encoding the Numbers

- $0 = \emptyset$
 - $1 = \{\emptyset\}$
 - $2 = \{\emptyset, \{\emptyset\}\}$
 - $n + 1 = n \cup \{n\}$
-
- $m < n \rightarrow m \in n \text{ and } m \subset n$
 - Basis of ZFC

Russell's Paradox

- $R = \{X : X \notin X\}$
- $R \in R?$
- If it is: $R \notin R !!!$
- $R \notin R$
- Then $R \in R !!!$